Approval-Based Elections in the 1D Euclidean Domain

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Chapter 1

Introduction

Elections are a ubiquitous element of our society in one form or another. For example, societies need to choose members of parliaments, judges have to pick winners of sports competitions. However, elections are not only applicable to politics or competitions. In more day-to-day examples of elections, teams have to schedule meetings times, or video streaming platforms need to provide recommended movies list to the users. Each of these problems may require a different type of an election and a different strategy of selecting the winner (or winners). The differences come from the underlying properties and objectives of the problem. For example, in presidential elections, we want to choose the candidate supported by the majority of the voters. It is not the case for the parliamentary elections, where the goal is to select members of the parliament in such a way that the opinions of the voters are represented proportionally. Another example would be a meeting scheduling problem, where we will try to pick the time, such that, we maximize the number of participants, even though, the selected time might not be optimal for anyone (e.g., early morning). In sports competitions, especially in the seasonal ones, i.e., the ones consisting of series of events, we need to come up with a strategy of calculating overall ranking at the end of the season, based on the aggregated results from the single events. For example, in FIS Alpine Ski World Cup, or in Formula 1 World Championship, the overall winners achieve very high scores consistently throughout the season, but might not have necessarily achieved the highest scores (1st place) in the majority of the individual events.

Another key distinction between the given examples, is whether a given election has just one winner, or does it lead to selecting multiple candidates. More formally, in computational social choice theory, these election types are called *single-winner* or *multiwinner* elections. Multiwinner (committee) elections, are even more widespread than the single-winner ones, and in most cases they have different properties and require different voting rules. For an overview of various committee election types and their related properties, see works

of Faliszewski et al. (2017) and Elkind et al. (2017). A committee election is the process of choosing a fixed-size subset of candidates (i.e., the committee), based on the voters' preferences. Voters' ballots (preferences) can be represented in various different ways. In this work we focus on approval elections, that is, on elections where each voter provides a subset of candidates that he or she approves of. Another notable type of elections is the ordinal model, where the voters order the candidates from the most to the least preferred ones. For an extensive survey of the approval model and current state of research in this area, regarding committee elections, we point to a survey by Lackner and Skowron (2020).

Winner determination is one of the core problems in computational social choice theory, however not the only one. Other ones include, e.g., control problems (Bartholdi et al. (1992); Hemaspaandra et al. (2009)), which refer to certain ways of manipulating elections. In short, we try to make a given candidate win (or lose) by changing the structure of an election, that is, by adding or deleting voters or candidates. Other interesting problems, that we are not going to explore in this work, include manipulation (Bartholdi et al. (1989); Faliszewski and Procaccia (2010)), bribery (Faliszewski et al. (2006)), justified representation (Aziz and Walsh (2014)) and many others (Brandt et al. (2016)).

1.1 Preference Restrictions

Many of the above-mentioned social choice problems are known to be computationally hard. We can deal with this complexity using various methods, such as designing approximation (random or deterministic) or parametrized algorithms. An alternative approach is to introduce restrictions on the preference domains. In other words, we restrict the structure of voters' ballots. There is a successful and active line of research in this area. This begins with the notion of *single-peaked (SP)* (Black (1958); Arrow (1951)) and *single-crossing (SC)* (Mirrlees (1971); Roberts (1977)). Much more recently, Barberà and Moreno (2011) introduced a concept of the *top-monotonicity*, a generalization of SP and SC restrictions. These models provided a basis for numerous and promising results in circumventing the computational complexity in a variety of problems. (Walsh (2007); Faliszewski et al. (2009); Skowron et al. (2015); Brandt et al. (2015); Elkind et al. (2016))

However, in the context of this work we need to emphasize a survey by Elkind and Lackner (2015). Unlike most of the above-mentioned examples, which considered the ordinal model of preferences, the survey explored well-known preference restrictions in the setting of approval preferences, and introduced several new ones. This includes models such as *Voter Interval (VI)* and *Candidate Interval (CI)*, both based on the *Dichotomous Euclidean (DE)* restriction. Euclidean models are very intuitive, a simplified explanation of such models,

is that, each candidate and each voter is assigned an ideal point in some Euclidean space $\mathbb{R}^d$, this point represents their opinion on some d issues. The points are assigned in such a way, that voters prefer the candidates that are closer to them. Godziszewski et al. (2021) extended this model, making it more versatile, by using a radius in addition to a point, to represent candidates and voters opinions. Godziszewski et al. (2021) named the new model a *Voter/Candidate Range (VCR)* model, and later analyzed the two-dimensional case and briefly the one-dimensional one, for committee election problems.

1.2 Contributions

In this work we focus on the one-dimensional Voter/Candidate Range model and our main contributions are as follows:

1. In Chapter 3, we provide detection algorithm for the 1D-VCR property, based on a reduction to integer linear programming (ILP). Additionally, we apply it to detecting 1D-VR and 1D-CR profiles, later we do a performance comparison with a solution based on a 2-SAT reduction from Fitzsimmons and Lackner (2019).

2. In Chapter 4, we analyze the 1D-VCR domain and compare it to other well known models. We use multiple statistical cultures to generate random elections in the 1D-VCR model, then we check how many of them do not satisfy 1D-VR or 1D-CR properties. Furthermore, we inspect how different are the 1D-VCR elections from the 1D-VR and 1D-CR ones.

3. In Chapter 5, we propose a polynomial time algorithm for a certain control problem under the 1D-VCR model, expanding on the solutions for 1D-CR and 1D-VR models, due to Magiera and Faliszewski (2014) and Faliszewski et al. (2009), respectively.

(WIP: Powyżej, używam VR i CR, mimo że nigdzie o nich nie wspomniałem, czy powinienem napisać że VI=CR, CI=VR?)

We do not discuss the related work right now, and instead we do it throughout this thesis, as the discussion will greatly benefit from the added context and introduced definitions in the subsequent chapters.

Chapter 2

Preliminaries

First, in Section 2.1, we introduce basic definitions and core concepts, together with simple examples. Later, in Section 2.2, we discuss preference restrictions, and how can we leverage them to design better algorithms, for many of the computational social choice problems. Finally, in Section 2.3, we explain some computational methods (tricks) we have used in this work, that we believe are not necessarily straightforward.

2.1 Basic Definitions

Approval Elections

An approval-based election (in short, an *election*, or interchangeably a *profile*) is a pair $E = (C, V)$, where $C = \{c_1, ..., c_M\}$ is a set of candidates and $V = \{v_1, ..., v_N\}$ is a set of voters. We use M and N for the number of candidates and voters respectively. For each voter $v \in V$, his or hers preferences are represented as a set of approved candidates $A(v)$, we call it an *approval set*. Furthermore, for each candidate $c \in C$, we define $V(c) = \{v \in V \mid c \in A(v)\}$, a set of voters who approve candidate c.

In practice, we often use an additional, simplified notation of representing an election. We redefine A as a two-dimensional binary matrix, with N rows and M columns, where value of $A[i][j]$ is (1) if i-th voter approves of j-th candidate, and (0) otherwise. In other words, each row of the matrix is a vector representation of a given voter's approval set. We refer to it as an *approval matrix* or a *preference matrix*. It will always be clear from the context which notation we are using.

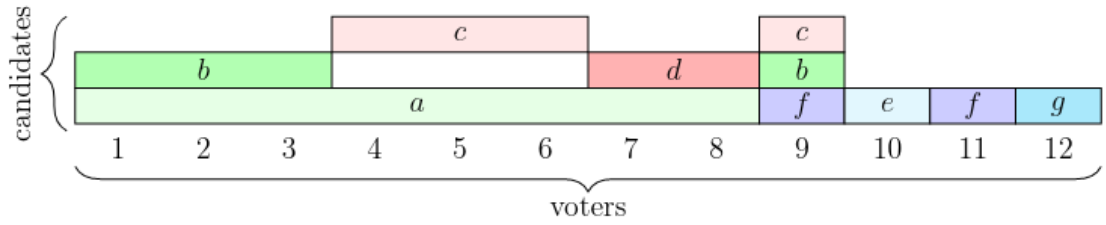

Figure 2.1 Temporary Image - Example of An Approval Election Lackner and Skowron (2020)

Example 1. *(WIP: Opis wyborów + figura) Following the example from Figure 2.1 ...*

Multiwinner Committee Selection Rules

Voting rules are algorithms used to, select a committee (winners) — fixed size subset of the candidates, based on the voters preferences, for a given election. Each rule has different properties and must be selected according to the goal of the election. As this is not the main focus of this work we point to Faliszewski et al. (2017), for a detailed overview of multiwinner voting rules.

Nonetheless, we introduce the most basic voting rule for the approval model, simply called *Approval Voting (AV)*, introduced by Brams and Fishburn (1983).

Definition 2.1.1 (Multi-Winner Approval Voting (AV))**.** *For a given election E and a size of committee k, AV rule selects the k candidates with the highest approval scores. AV-score of a given candidate c, is the number of voters who approve of c, and formally, is defined as* $score_{AV}(E,c) = |\{v \in V : c \in A(v)\}|$.

Example 2. *(WIP: Opis wyborów z poprzedniego przykładu + kto wygra) Continuing with the election from Example 1*

2.2 Preference Restrictions

As a reminder from Section 1.1, many computational social choice problems such as winner determination, or control problems are known to be computationally hard. Preference restrictions, i.e., restrictions on the structure of voters' ballots, is a very effective solution to circumvent the computational complexity of these problems.

2.2.1 Euclidean Preference Restrictions

Euclidean preference restrictions, in the approval voting model, are foundation of this work. All the problems we solve in the subsequent chapters are structured around this concept. Unless stated otherwise, throughout this work, we consider one-dimensional variants of the Euclidean models.

First, the most basic restriction, that is an important building block to describe other ones, in this family of restrictions.

Dichotomous Euclidean (DE)

(Elkind and Lackner (2015)) Given an election $E = (C,V)$, we say that E satisfies *DE* if there is a mapping ρ of voters and candidates into the real line such that for every voter $v \in V$ there exists a radius r_v, and it holds that $v = \{c : |\rho(v) - \rho(c)| \leq r_v\}$.

(WIP: Czy w ogóle definiować DE?, czy jest potrzebna cytacja?, to jest definicja z Elkind and Lackner (2015) nie jest specjalnie oczywista)

Voter/Candidate Range (VCR)

(Godziszewski et al. (2021)) Given an election $E = (C,V)$, we say that the voters have *one-dimensional Euclidean* preferences (VCR) if for every agent $a \in C \cup V$ (i.e., for each candidate and each voter) there exists a point x_a and nonnegative real value radius r_a such that:

$$c \in A(v) \iff |x_c - x_v| \leq r_c + r_v$$

We can easily think of it as, for an agent $a \in C \cup V$ the point x_a describes a's ideal position in $\mathbb{R}$, representing his or hers opinion. For a candidate $c \in C$, radius r_c can be seen as c's charisma, or in other words their ability to reach a number of different voters (e.g., a centrist candidate would have a larger reach, than an extremist one (left or right)) For a voter $v \in V$, radius r_v corresponds to v's willingness to compromise, that is, how different (faraway) opinion they would still accept from their ideal one.

Moreover, there are two special cases of VCR model.

1. *Candidate Range (CR)*, we say that a VCR election $E = (C,V)$, satisfies CR if for each voter $v \in V$, their radius r_v is equal to zero.

2. *Voter Range (VR)*, we say that a VCR election $E = (C,V)$, satisfies VR if for each candidate $c \in C$, their radius r_c is equal to zero.

The CR and VR models are equivalent to the *voter interval (VI)* and *candidate interval (CI)* ones, both introduced by Elkind and Lackner (2015). Godziszewski et al. (2021) simply use a different terminology.

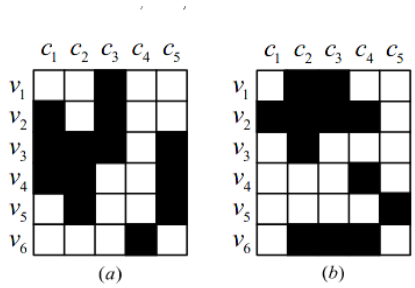

Figure 2.2 CR and VR Temporary image from Liu and Guo (2016)

The VCR domain is larger than the CR and VR domains combined. Below in Example 3, we show an election that is VCR, but neither CR nor VR. We explore this topic further in Chapter 4.

Example 3. *Given an election E with the candidates $C = \{a,b,c,d\}$ and the voter's approval sets $\{a,b\}$, $\{b,c\}$, $\{b,d\}$, and $\{a,b,c,d\}$.*

To see why, ... (WIP: Podać punkty i promienie (VCR), następnie opisać dlaczego nie jest CR ani VR ... Obrazek?)

2.3 WIP: Methodology/Techniques

2.3.1 Integer Linear Program

Integer linear programming (ILP) is a common approach to solve NP-hard problems, furthermore it can be used effectively for other problems as well. (WIP: czy zdanie od furthermore ... jest potrzebne) To define a computational problem as an integer linear program, we have to create a system of linear inequalities (constraints) (WIP: Sprawdzic czy na pewno inequalities == constraints) and optionally a linear function to optimize, alternatively we can just check feasibility of the constraints. Next to obtain a solution we have to use an optimization solver. Popular choices are Gurobi or CPLEX which are state-of-the-art general purpose solvers.

Chapter 3

Voter/Candidate Range Profile Detection

In this chapter we present our algorithms for deciding if a given approval profile has the VCR property.

3.1 Profile Structure Detection

To take advantage of the restrictions defined in Section **??**, we need to design detection algorithms. They can be used to check whether given election profiles satisfy particular structural properties. For example, to check the VR (CR) property it is enough to check for the *consecutive ones property* in the preference matrix.

Definition 3.1.1. *A binary matrix (i.e., a matrix consisting of zeros and ones) has the consecutive ones property (C1P) if its columns be permuted in such a way that for each row the ones are consecutive, i.e., the 1s form an interval.*

In this work we refer to the term *C1P* more broadly, using it interchangeably, for an analogous scenario where we permute the rows and look for the consecutive 1s in every column in a given matrix. Booth and Lueker (1976) provided a linear-time C1P detection algorithm based on the PQ-Tree data structure.

Proposition 1 (Elkind and Lackner (2015))**.** *Deciding whether an approval profile satisfies VR or CR is possible in $O(m \cdot n)$ time.*

Proof. Let $C = \{c_1, ..., c_m\}$ and $V = \{v_1, ..., v_n\}$, we create a $n \times m$ matrix A (a_{ij}): for each $i \in [n]$ and $j \in [m]$: $a_{ij} = 1$ if $c_j \in A(v_i)$ and 0 otherwise. We can see that for the VR property, permuting the columns of matrix A so that ones form an interval in each row is equivalent to permuting candidates so that the set of candidates approved by each voter forms an interval. For the CR case, we just need to transpose the matrix created for the VR. □

Our focus lies on the one dimensional VCR property. Having preference matrix as input we have to provide for each agent (voter/candidate) a point on the real line and a nonnegative real value representing radius, so that these parameters satisfy the VCR property.

3.2 Integer Linear Program

An algorithm based on ILP solving is an excellent choice for the first attempt at designing a detection algorithm. It provides an efficient and relatively simple solution. Furthermore, it can be used as a fast benchmark for validating future solutions. Below we provide an ILP for the VCR profile recognition, together with additional preliminary steps.

Transforming the Definition of VCR into Linear Constraints

Before formulating an ILP for detecting VCR profiles, we have to redefine the definition of VCR in order to make it appropriate for an ILP solver. The definition is based on the euclidean distance which cannot be directly represented in an ILP. Equation 3.2 shows transformation of the VCR definition into linear inequalities.

As a short reminder from Section 2.2.1, for the candidates C and voters V, 1D-VCR is defined as:

$$\forall\, c \in C\, \forall\, v \in V : c \in A(v) \Leftrightarrow |x_c - x_v| \leq r_c + r_v \tag{3.1}$$

Where x_a and r_a represent agent's point on the real line and their radius respectively.

(WIP: W jaki sposób mogę zwięźle, bez ponownego wpropadzania wszystkich oznaczeń, przypomieć definicję? O ile w ogóle jest to potrzebne...)

$$\begin{aligned} &|x_c - x_v| \leq r_c + r_v \\ &\iff -(r_c + r_v) \leq x_c - x_v \leq r_c + r_v \\ &\iff x_c - x_v \leq r_c + r_v \;\; \vee \;\; -(x_c - x_v) \leq r_c + r_v \end{aligned} \tag{3.2}$$

(WIP: Poprawić formatowanie, jak?)

The constraint obtained from the equation 3.2 is not suitable yet for an ILP because it contains logical *OR* condition which again cannot be directly represented in an ILP. To transform it into linear form, we apply commonly called *BigM* technique, as shown below.

$$
\begin{aligned}
x_c - x_v &\leq r_c + r_v + M(1 - z_1) \\
-(x_c - x_v) &\leq r_c + r_v + M(1 - z_2) \\
z_1 + z_2 &\geq 1 \\
z_1, z_2 &\in \{0, 1\}
\end{aligned}
\tag{3.3}
$$

First, we have to introduce a large constant M (larger than $max(|x|)$) and two binary variables z_1 and z_2. Now if any of the z_i is 0 then the corresponding inequality will be true, and the additional constraint $(z_1 + z_2 \geq 1)$ guarantees that at least one of the z_i will be 1.

(WIP: Trochę namieszane wytłumaczenie, bo tak naprawdę problem z *OR* jest tylko dla wariantu if $c \notin A(v)$, czyli $x_c - x_v + M(1 - z) \geq r_c + r_v + 1$ ...)

The BigM technique is common enough, for most solvers to provide additional built-in mechanisms for applying it. Using them is usually recommended, as it allows a solver for better internal optimization. In Gurobi solver we use *indicator constraint*, which allows one to create versatile constraints in form of: $z = f \rightarrow a^T x \leq b$. The constraint states that if the binary indicator variable z is equal to f, where $f \in \{0, 1\}$, then the linear constraint $a^T x \leq b$, where a and b are data coefficients vectors and x represents the decision variables, should hold. On the other hand, if $z = 1 - f$, the linear constraint may be violated.

(WIP: BigM jest używane do reprezentacji różnych 'funkcji', nie tylko *OR*, np. *max*, *abs*, etc., czy też o tym wspomnieć?)

Detection Algorithm

Final formulation of the ILP for recognising VCR property in a given election profile is shown in Figure 3.4. The program accepts preference matrix A as input, and if solution exists it outputs every agent's position x_a and radius r_a. Additionally, we use two binary variables p and q, required by the indicator constraints.

$$
\begin{aligned}
&\textbf{check feasibility}: \\
&\quad \textbf{subject to}: \\
&\qquad x_a \in \mathbb{R} && \forall\, \mathrm{a} \in C \cup V \\
&\qquad r_a \in \mathbb{R}^{0+} && \forall\, \mathrm{a} \in C \cup V \\
&\qquad p_{cv} \in \{0,1\} && \forall\, \mathrm{c} \in \mathrm{C}\, \forall\, \mathrm{v} \in V \\
&\qquad q_{cv} \in \{0,1\} && \forall\, \mathrm{c} \in \mathrm{C}\, \forall\, \mathrm{v} \in V \\
&\quad \text{if } c \in A(v): \\
&\qquad x_c - x_v \leq r_c + r_v \\
&\qquad -(x_c - x_v) \leq r_c + r_v \\
&\quad \text{if } c \notin A(v): \\
&\qquad p_{cv} == 1 \rightarrow x_c - x_v \geq r_c + r_v + 1 \\
&\qquad q_{cv} == 1 \rightarrow -(x_c - x_v) \geq r_c + r_v + 1 \\
&\qquad p_{cv} + q_{cv} >= 1
\end{aligned}
\tag{3.4}
$$

3.3 WIP: Consecutive Ones Detection — ILP vs SAT

Chapter 4

Voter/Candidate Range Profiles

In this section we explore the Voter/Candidate Range domain and describe some of its properties. The main goal is to compare the size of the domain with other well known domains such as Voter Range and Candidate Range.

4.1 Domain analysis

WIP: Najpierw Napisać dlaczego / po co to robimy. Dopiero potem co / w jaki sposób. Chcemy sprawdzić czy warto wymyślać algorytmy VCR i czy doemna jest znacząco większa od VR / CR

We have performed several experiments, in each of them we check how many profiles have the VCR property but not VR neither CR. For the first one we focus on the small size elections, and for the second experiment we employ several strategies to generate random, larger profiles.

4.2 How many VCR profiles are not VR neither CR?

Definition 4.2.1. *A profile P is **Truly VCR (TVCR)**, only if P satisfies VCR property and does not satisfy VR and CR properties.*

Definition 4.2.2. *A profile P is **Fully COP (FCOP)**, only if P satisfies VR and CR properties.*

4.2.1 Experiment 1 — Small Profiles

For the small profiles we did a complete search of the domain. That is we generated every possible profile of a given size and checked if it has the VCR property. We checked 6

dimensions of the elections with the following sizes, having M candidates and N voters, $(M,N) \in \{(3,3),(4,4),(5,5),(4,6),(6,4)\}$. The results as seen in Figure 4.1 show that VCR domain is relatively large, however, for these small election sizes, only limited number of profiles are Truly VCR (i.e., neither VR nor CR).

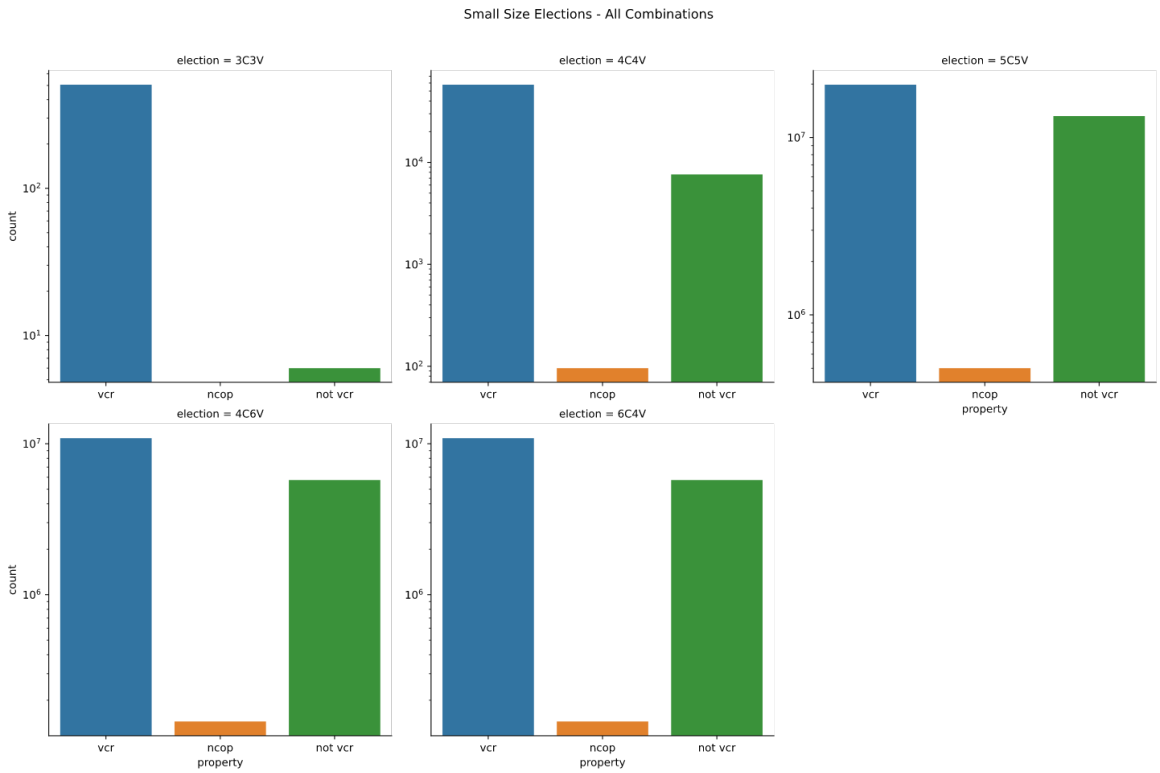

Figure 4.1 Temporary Image - Aggregated Statistics For Small Profiles

4.2.2 Experiment 2 — Large Profiles

For the larger size of the profiles we generated a suitable number of random VCR profiles and again checked how many of them are neither CR nor VR (*TVCR*), and additionally, VR and CR (*FCOP*), VR and not CR, CR and not VR. We employed several strategies for generating the profiles. The first parameterized category is a statistical distribution used to pick the points representing voters and candidates. The second one is the choice of radius parameter. We describe the categories in detail below. For each parameter pair (in total 20 pairs) we generated 1000 elections with 20 candidates and 20 voters, then we repeated the experiment for the profiles consisting of 40 candidates and 40 voters.

Position Parameter

We consider 4 ways of generating the points, representing the positions of the candidates and the voters.

1. The *Uniform Candidates, Uniform Voters* model, where the points for both, the candidates and the voters are selected uniformly at random from the $[-10, 10]$ range.

2. The *Asymmetric Gauss Candidates, Asymmetric Gauss Voters* model, where we take 70% of points from a Gaussian distribution with mean value -3 and standard deviation 1.8, and the remaining 30% come from a Gaussian with mean 3 and the same standard deviation. This model is inspired by the 2 dimensional model from the work of Godziszewski et al. (2021). The goal is to simulate the society with a large majority, and a minority with the opposite opinion.

3. The *Uniform Candidates, Asymmetric Gauss Voters* model, it is a combination of the previous two models, where the candidate's points are selected uniformly at random, and for the voters we use asymmetric Gaussian model. We use the same parameters for both of the distributions.

4. The *Asymmetric Gauss Candidates, Uniform Voters* model, it is mirror version of the previous model, with the distributions swapped.

Radius Parameter

In this category we consider 5 ways of generating the radii. We split the methods between the candidates and the voters, similarly to picking the positions.

1. The *Gauss Candidates, Gauss Voters* model, where we choose the radii from the Gaussian distribution with mean value 1.5 and standard deviation 0.5 We use the same distribution for the candidates and the voters.

2. The *Small Uniform Candidates, Gauss Voters* model. We pick radii for the candidates uniformly at random from the range $[0.7, 1.2]$, and for the voters we use the same Gaussian, as in the previous model ($Gauss(1.5, 0.5)$).

3. The *Gauss Candidates, Small Uniform Voters* model. In this model we swap the distributions used in the previous one, that is, the candidates are picked from the Gaussian and the voters from the uniform distributions.

4. The *Small Uniform Candidates, Small Uniform Voters* model. We pick radii for both the candidates and the voters, uniformly at random from the range $[0.7, 1.2]$.

5. The *Large Uniform Candidates, Large Uniform Voters* model. We pick radii for both the candidates and the voters, uniformly at random from the range $[0, 3]$.

At first, it might be surprising why we don't include a constant radius, as one of the parameters. That is because, then all the profiles would be VR and CR (*FCOP*), as we show in Proposition 2.

Proposition 2. *If a profile P satisfies VCR and every agent (candidate/voter) in the profile, has the same constant radius, then P satisfies both VR and CR properties.*

Proof. Because. (WIP: Dokończyć, R=const –> 2 razy większy R kandydata i R=0 głosujacego, i vice versa) □

Results

First, we present the visualization of our example election profiles, and how their visual structure is impacted by the generation parameters in Figure 4.2. Later we analyze aggregated results of the experiment. Most of the conclusions we draw are inline with the basic intuition.

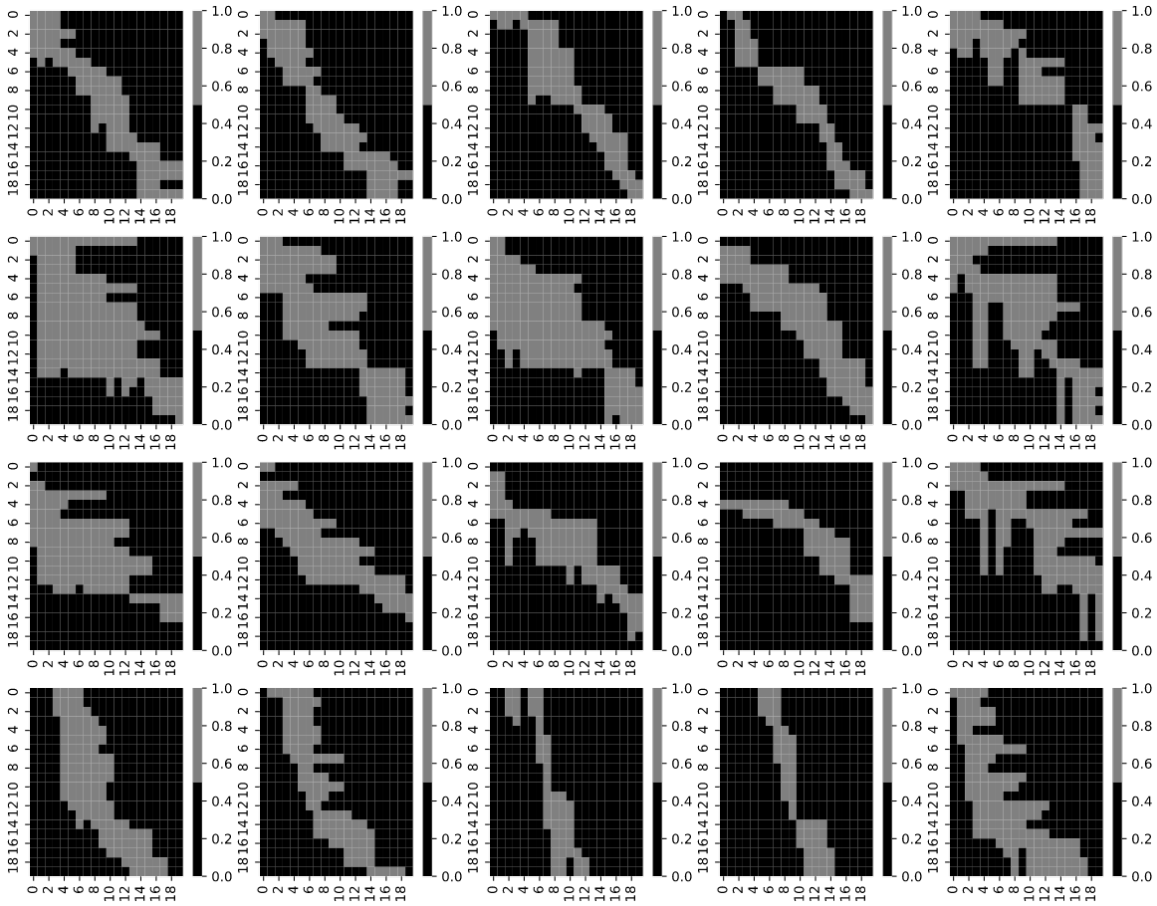

Figure 4.2 Temporary Image - Cols= R, Rows = X

Election Dimensions Our results for two election sizes have one crucial difference, looking briefly at Figure 4.3 (20 candidates, 20 voters) and Figure 4.4 (40 candidates, 40 voters), we immediately see, that in the larger elections *TVCR* property is much more frequent. (WIP: Napisać jakąś hipotezę, dlaczego tak się dzieje?) Election sizes in the experiment are relatively small when compared to the real world scenarios. We can expect for this trend of increased number of *TVCR* profiles to continue for even larger elections. This result is very promising, as it strongly encourages further development and research related to the VCR domain.

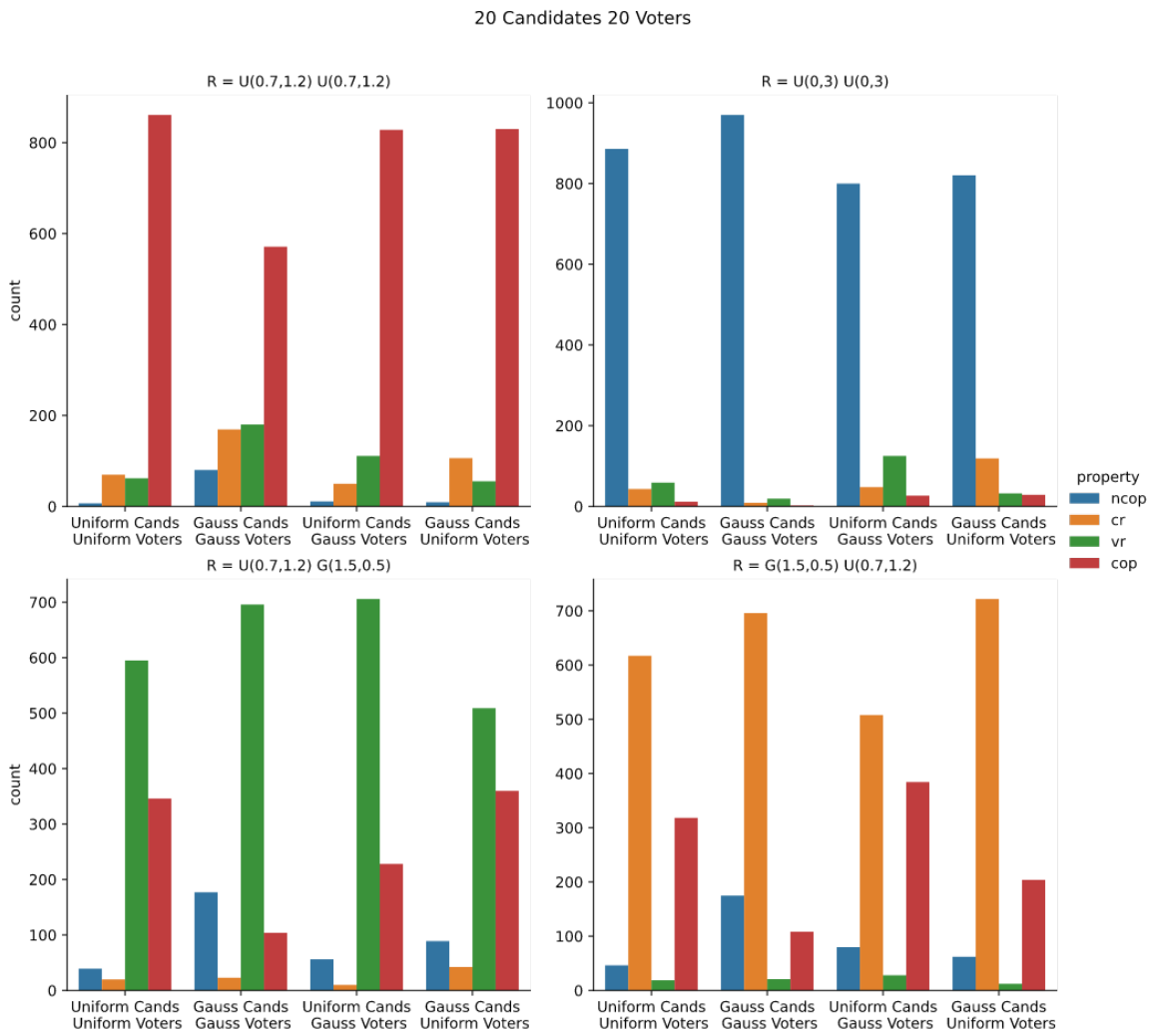

Figure 4.3 Temporary Image - 20C 20V Aggregated Statistics

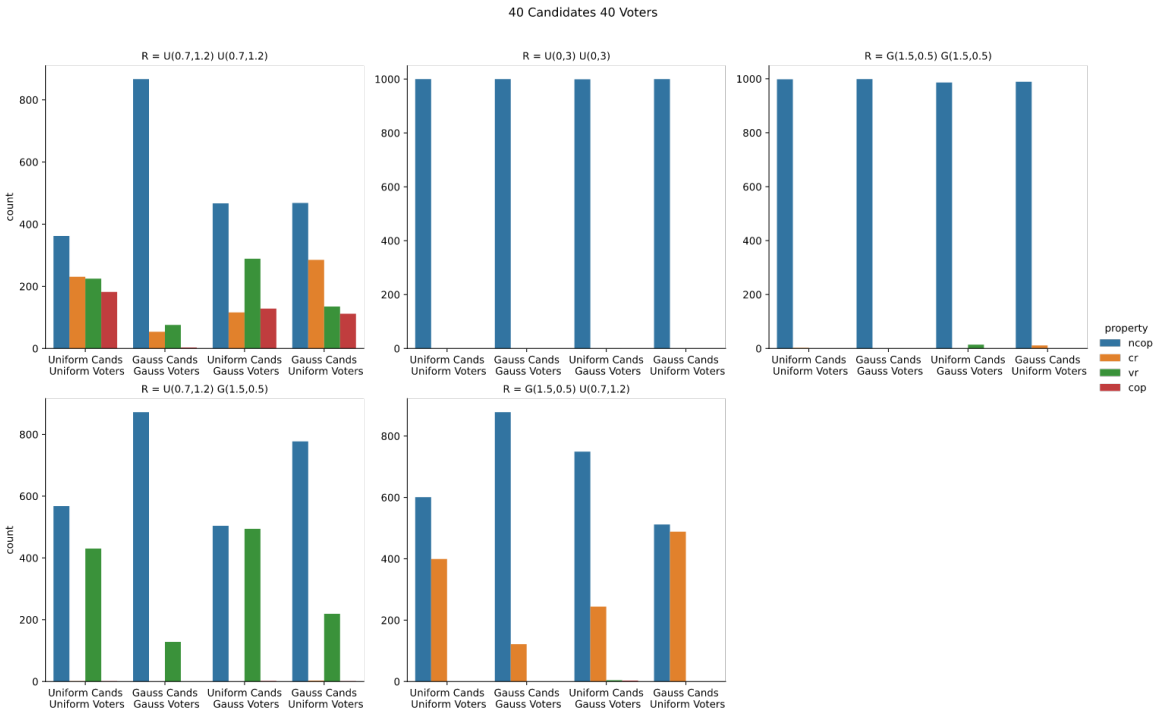

Figure 4.4 Temporary Image - 40C 40V Aggregated Statistics

In the subsequent paragraphs we will focus mostly on the results for smaller elections from Figure 4.3, but only because the differences between the parameters are more pronounced.

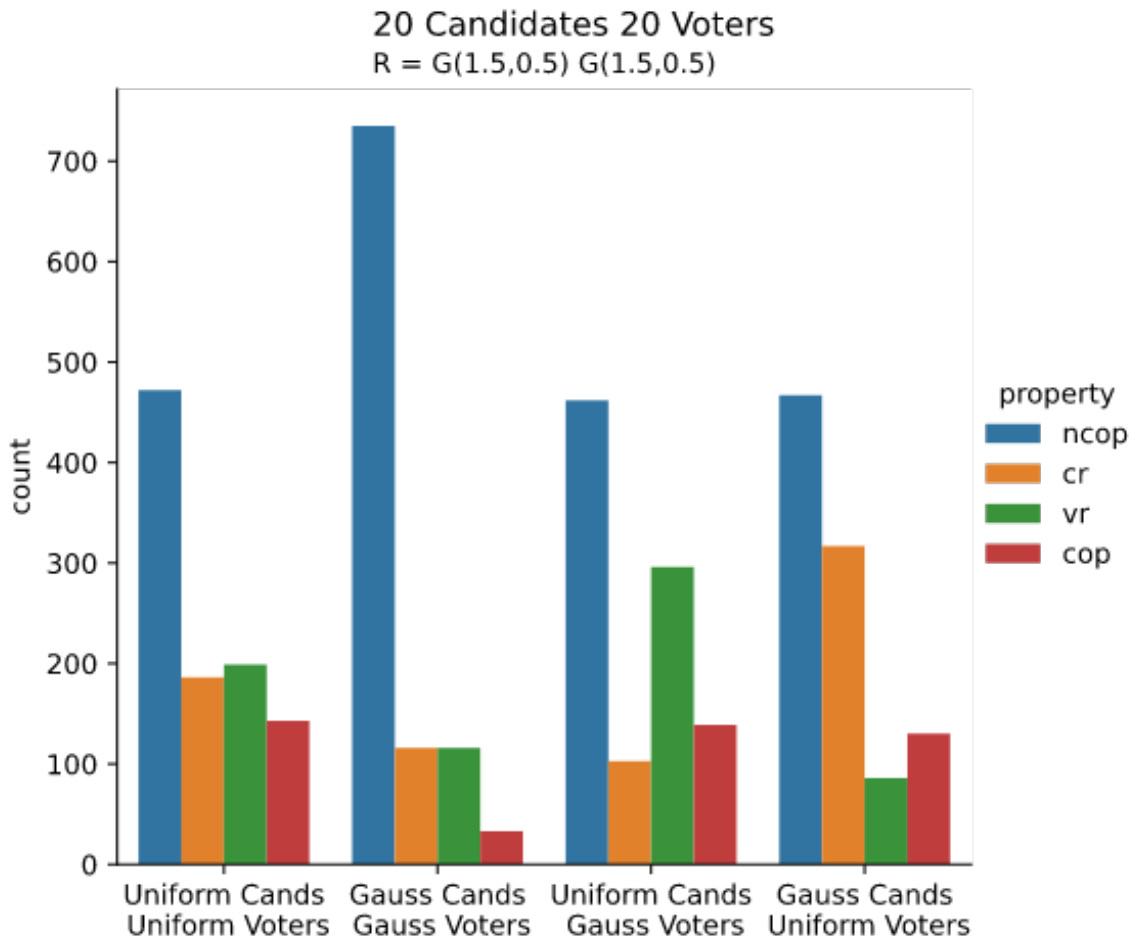

Figure 4.5 Temporary Image - Selected Histogram - Double Gauss Radius Aggregated Statistics

Position Choice We will analyze Figure 4.5 — histogram for the ***Gauss Candidates, Gauss Voters*** radius model, as this model exhibits the most neutral behavior. It will be our baseline scenario. We can compare position models in pairs. Starting with *Uniform Candidates, Uniform Voters* and *Asymmetric Gauss Candidates, Asymmetric Gauss Voters* models. The double Gauss model has noticeably more *TVCR* profiles. It can be easily explained by looking at the respective preference matrix heatmaps in Figure 4.2. The profiles generated from the double Gauss model have higher count of approvals, which in turn increases the likelihood of any irregularities in the approvals block structure, which is directly linked to the *TVCR* property. (WIP: lepiej wyjaśnić, mam na myśli "dziury" w blokach jedynek w macierzy)

Now turning our attention to the second pair of the position models (still for the same radius model), *Uniform Candidates, Asymmetric Gauss Voters* and *Asymmetric Gauss Can-*

didates, Uniform Voters. These are two perfectly symmetrical models, we will explain only the first one, and keep in mind that the second one is simply a reversed scenario. Looking at the heatmaps again, we notice that the blocks of approvals are skewed horizontally, which is associated with the VR property, i.e., the ones form and interval in each row of the approval matrix. On the contrary, in the second model, the blocks are skewed vertically, which links to the CR property.

Radius Choice From Figure 4.3 it is clear that the choice of radius can significantly reinforce or subside trends (in the profiles properties statistics) resulting from the position choice. For example, the mixed radius models, i.e., *Small Uniform Candidates, Gauss Voters* and *Gauss Candidates, Small Uniform Voters*. In particular, these models increase occurrence of profiles with VR or CR properties. The core idea behind this phenomenon is exactly the same as the one described above, for the mixed position models (Uniform, Gauss), where the blocks of approvals are skewed, either horizontally or vertically, resulting in the increased number of VR or CR profiles respectively.

The striking difference between histograms for *Small Uniform Candidates, Small Uniform Voters* and *Large Uniform Candidates, Large Uniform Voters* is again related to the inconsistencies in the approval blocks, which are easily observed in the heatmaps. Smaller radius range produces much more structured profiles, hence large number of the *FCOP* profiles. In contrast, large radius range outputs irregular approval blocks in the profiles, increasing number of the *TVCR* profiles.

Summary The results are very promising, we showed that the *VCR* domain can be significantly larger than *VR* and *CR* combined. We hope, this will encourage further development and research related to the VCR domain.

4.3 Transforming a Truly VCR Profile into VR or CR one

A natural question that arises from the previous experiment results, is how different are the TVCR profiles from the VR or CR ones? In other words, how much do we need to change a TVCR profile for it to become a VR or CR one. To answer this question, we design another experiment.

4.3.1 Experiment 3 — Profile Transformation

In this experiment, we partially reuse profiles generated for the previous one. We narrow down our experiment parameters to only two radius generation methods, that is, first model *Gauss Candidates, Gauss Voters* and second *Large Uniform Candidates, Large Uniform Voters*. We keep the position models unchanged, and we explore all four of them. Obviously we only pick the TVCR profiles, to be exact, we use 400 of them for each parameter pair. Due to limited computational resources, we chose to only use the profiles consisting of 20 candidates and 20 voters.

Agent Deletion Parameter We consider 3 methods to transform a TVCR profile into a VR or CR one.

1. The *Candidate Deletion* method, where we search for a minimal number d, such that, after deleting d candidates from the profile it becomes a VR or CR profile.

2. The *Voter Deletion* method, which is similar to the *Candidate Deletion* method, but this time we delete d voters instead.

3. The *Candidate and Voter Deletion* method, where we combine the two previous methods, and we simultaneously delete d_c candidates and d_v voters.

Transformation Result Parameter Apart from choosing what type of agent (candidate / voter) to delete, for each profile we collect the transformation results into 3 groups. By the transformation result we mean the lowest possible cost of the transformation, i.e., the minimal number of agent deletions for a TVCR profile to become a VR or CR one.

1. The *Candidate Range*, where the cost is a number of deletions required to get a CR profile.

2. The *Voter Range*, where the cost is a number of deletions required to get a VR profile.

3. The *Candidate Range or Voter Range*, where the cost is a minimum from the previous two groups.

Results

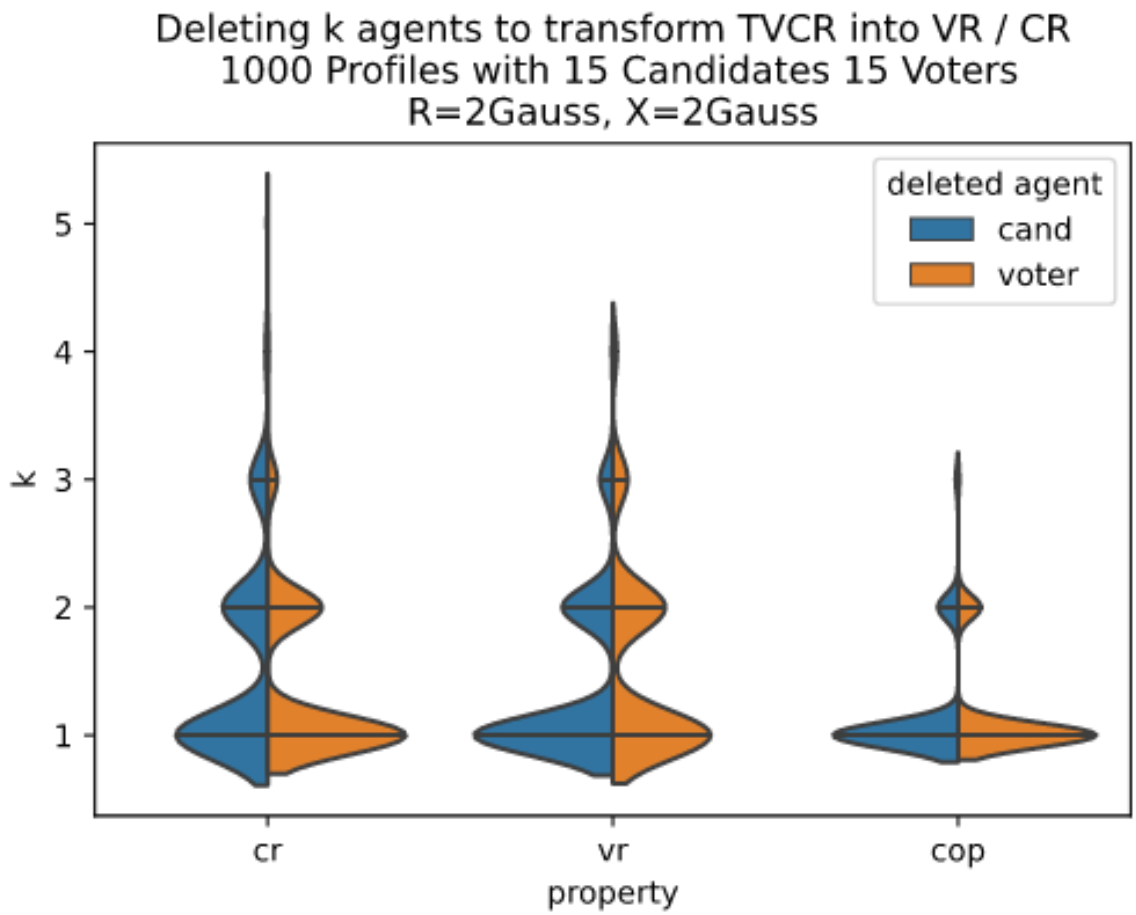

Figure 4.6 Temporary Image - Transformation of TVCR into VR/CR profile

Chapter 5

Control Problem

5.1 Overview

5.1.1 AV-CC-DV for Single Peaked (VR = CI = row C1P = $r_c = 0$)

Algorithm from Faliszewski et al. (2009)

1. kandydat p na osi, pozostali na lewo i prawo
2. wybór przeciwników (groźny kandydat):
   (a) pierwszy na lewo c_L i pierwszy na prawo c_R takich, że $score(c_?) \geq score(p)$
   (b) powtarzamy dla kolejnych lewych c_{L_i} takich, że $score(c_{L_i}) > score(c_{L_{i-1}})$
   (c) analogicznie dla prawych c_{R_i} takich, że $score(c_{R_i}) > score(c_{R_{i-1}})$
3. usuwanie głosów - zaczynając od najbliższego prawego przeciwnika c_R (dla lewych - odbicie lustrzane)
   (a) notacja v: $b(v)$ początek przedziału (lewy) i $e(v)$ koniec przedziały (prawy)
   (b) wybieramy głosujących $\widehat{V}$ takich że $\{v \in V | \, b(v) \in (p, c_R]\}$
   (c) sortujemy $\widehat{V}$ malejąco po $e(v)$ (od najbardziej prawego końca przedziału)
   (d) usuwamy $scoreDelta(p, c_R) + 1$ głosów
4. powtarzamy do skutu, przeliczając aprobaty co iterację

5.1.2 AV-CC-DV for Candidate Range = VI = column C1P = $r_v = 0$

Algorithm from Magiera and Faliszewski (2014)

1. notacja CR dla c: $v_{f(c)}$ początek przedziału (lewy/górny) i $v_{e(c)}$ koniec przedziału (prawy/dolny)
2. kandydat p
3. Głosujący - wybieramy głosujących $V' = \{v \in V | \, p \notin A(v)\}$
4. Przeciwnicy - wybieramy kandydatów $\widehat{C} = \{c \in C \setminus \{p\} | \, scoreDelta(c, p) \geq 0\}$
5. Usuwanie głosów
   (a) zaczynając od przeciwnika c_N takiego, że $v_{e(c_N)}$ jest minimalne
   (b) usuwamy $deltaScore(c_N, p) + 1$ głosów, zaczynając od końca przedziału $v_{e(c_N)}$
6. powtarzamy do skutu, przeliczając V' i $\widehat{C}$ co iterację

5.2 AV-CC-DV for the VCR profiles

Notation used in the algorithm and later in the proofs:
x_a, r_a — point and radius
$b(a) = x_a - r_a$ — range beginning (left)
$e(a) = x_a + r_a$ — range end (right)
p — candidate chosen to be a winner
$\widehat{C} = \{c \in C \setminus \{p\} | \, scoreDelta(c, p) \geq 0\}$ — opponents (dangerous candidates)
$V' = \{v \in V | \, p \notin A(v)\}$ — potential voters to delete

We use “smart” greedy strategy for voter deletion:

1. We pick a leftmost opponent c_L that is the one with the minimal $b(c_L)$
2. We consider only voters who approve of c_L that is, $\widehat{V} = \{v \in V' | \, c_L \in A(v)\}$
3. We need to delete $deltaScore(c_L, p) + 1$ rightmost voters from $\widehat{V}$ that is the ones with the maximal $e(v)$, if it’s not possible we fail
4. We repeat the above steps until we succeed, recalculating $\widehat{C}$ and $\widehat{V}$ after each iteration

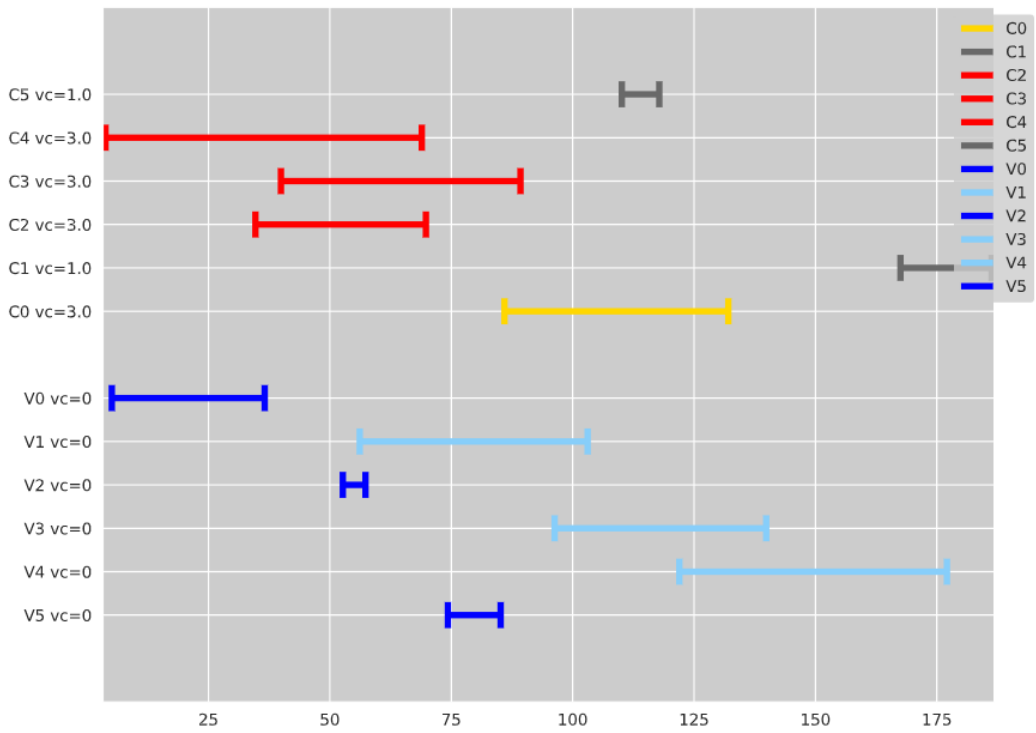

Figure 5.1 Temporary Image - VCR !C1P Election
yellow — candidate p
red — opponents
lightblue — voters whitelist
darkblue — possible voters to delete
grey — agents that can be ignored

5.2.1 Sketch Proof - CR case

We will show how the VCR algorithm is equivalent to the CR one from Magiera and Faliszewski (2014). Having voters in CR order, that is, they are still represented by a point x_v on the $\mathbb{R}$ line but have radius 0. Example in Figure 5.2.

Choosing opponents

In the CR algorithm, we choose candidates to process, ordered by $v_{e(c)}$ ascending. We can see that it is equivalent to picking the leftmost $(min(b(c)))$ candidate from the set of opponents $\widehat{C}$ in the VCR algorithm, because it will also have the minimal $v_{e(c)}$.

There is a special case of "contained candidate", for which the above statement is not true. In Section 5.2.3, we explain in detail why it is irrelevant for the overall correctness of the algorithm.

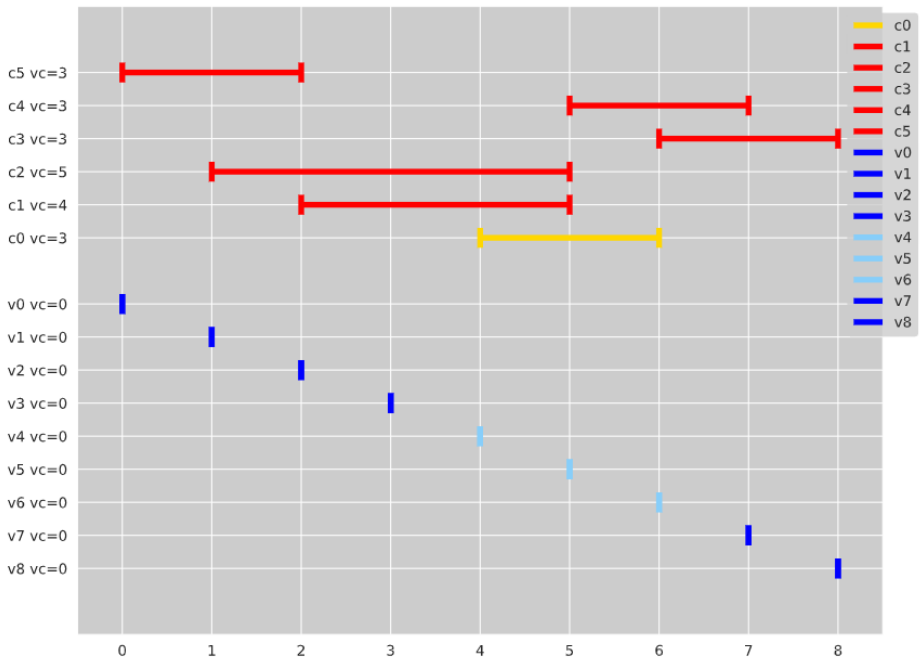

Figure 5.2 Temporary Image - CR Election

Voter deletion

Now, once we now how to pick an opponent c_L, we will show that choosing voters is equivalent as well. Notion of the beginning/end of the interval in the CR order is equivalent to the left/right end of the range in the VCR setting. Both algorithms try to delete $scoreDelta(p, c_L) + 1$ voters, CR one starts deleting voters from the end of interval, while VCR does virtually the same by deleting the rightmost voter first ($max(e(v))$).

5.2.2 Sketch Proof - VR case

We will show how the VCR algorithm is equivalent to the VR one from Faliszewski et al. (2009). Having candidates in VR order, that is, they are still represented by a point x_c on the $\mathbb{R}$ line but have radius 0. Example in Figure 5.3.

Choosing opponents

VR algorithm picks dangerous candidates to the left and right of candidate p, however later it proceeds only with defeating the right ones. Once finished with them, it does the "mirror flip" of the universe to process candidates on the left side. Even though the VCR algorithm picks leftmost candidate, we will show it is equivalent to the VR strategy under some basic assumptions.

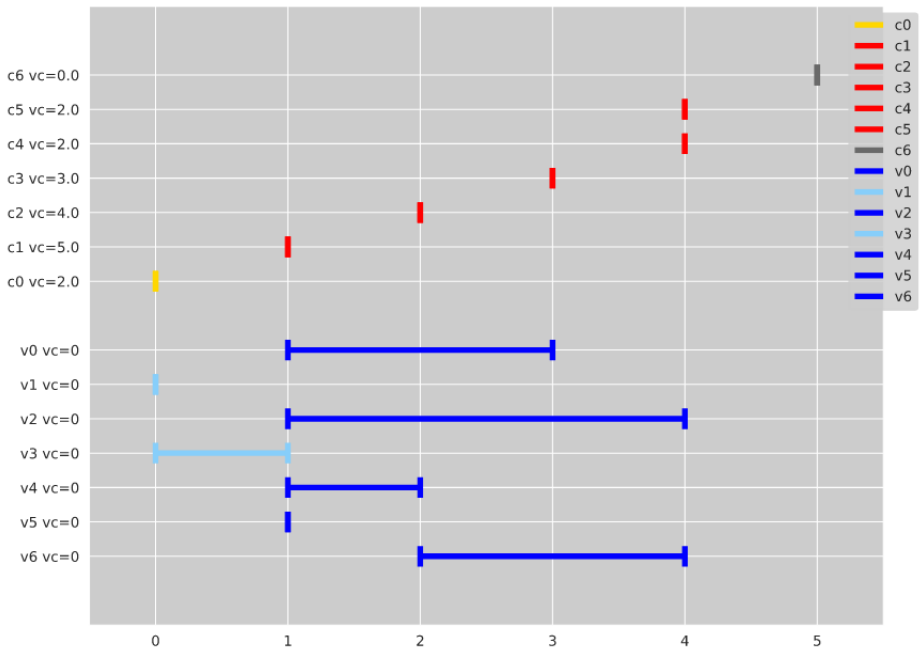

Figure 5.3 Temporary Image - VR Election

Following example in Figure 5.3, let's assume that candidate p always will be the first candidate in the VR order, which is equivalent to being the leftmost in the VCR order. Then we can see that the VCR algorithm strategy of choosing the leftmost candidate as an opponent will be the same as the VR algorithm choosing the first candidate on the right side of p. In the example it would be candidate c_1.

This assumption is correct because, having candidates in order $[c_L, p, c_R]$, if there was a voter v who approved of c_L and c_R it would also have to approve of p (because of the VR property), so we could not delete v anyway.

Now, expanding on the above, ...

(WIP: Faktycznie taki dowód nie jest w 100% przekonujący Mam 2 możliwości: 1) Wprowadzić krork "universe flip" do algorytmu i wtedy na pewno działa. 2) Zmienić argumentację i pokazać, że wybieranie najdalszego kandydat od p (najbardziej lewy po lewej stronie) też jest w porządku)

Voter deletion

The reasoning here is very similar to the one in the voter deletion part of the CR proof. Both algorithms try to delete $scoreDelta(c, p) + 1$ voters. VR algorithm starts deleting voters from the end of interval $(max(e(v)))$. VCR one does virtually the same by deleting the rightmost voter first.

5.2.3 Sketch Proof - VCR (not C1P) case

Now we will show why our solution is correct in the general VCR setting, even when the profile does not have VR/CR property, example in Figure 5.1. The proof is split into two stages, similarly to the previous ones. But first we will explain one special case of the election, that is worth examining.

Special case - contained candidates - alternative explanation

In the VCR model it is possible that some candidates have their interval fully contained in the other's interval. Every voter who would approve the smaller candidate would also approve the larger one, however not vice versa. For this case, our strategy of choosing voters to delete might seem suboptimal, at first glance.

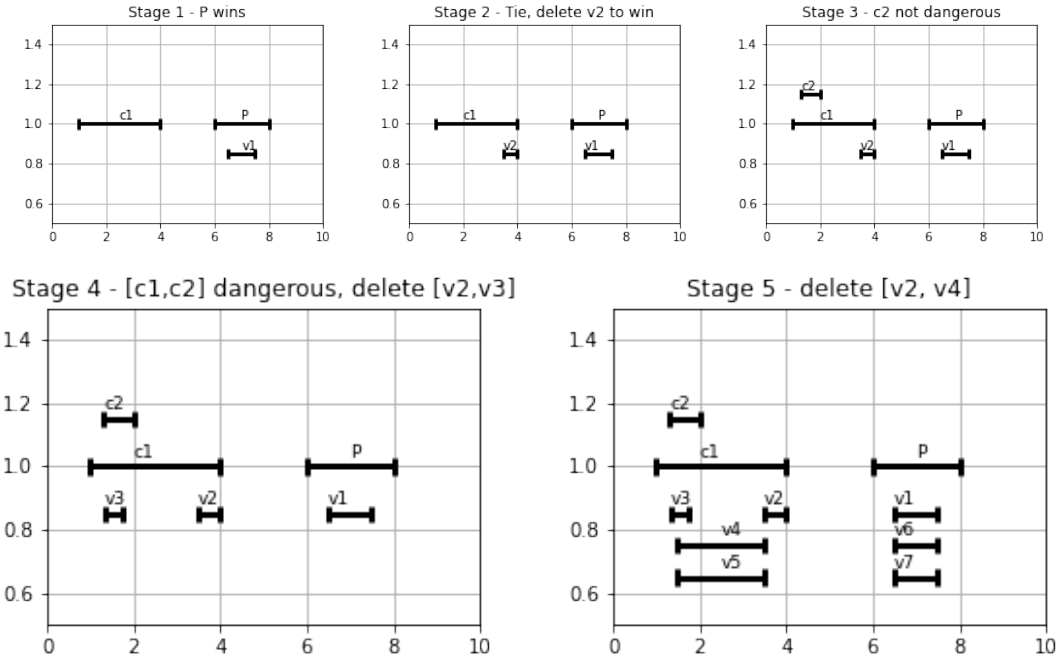

Figure 5.4 Temporary Image - Special Case

We explain the issue, starting with the example of the most basic election in Figure 5.4. Stages 1 through 3 are trivial, in the subsequent stages we make the election more and more complex by adding new agents. Turning our attention to Stage 4, now both candidates c_1 and c_2 are dangerous, with scores 2 and 1 respectively. Candidate p needs to defeat c_2 by one vote, and c_1 by two votes. In Stage 5 we add multiple more voters ...

The core idea to solve our issue is understanding what types of the new votes we can add. Let's consider only the voters who approve of c_1 and/or c_2.

1. A voter who approves only c_1 and not c_2, in this case, $scoreDelta(p, c_1)$ would increase by 1, and $scoreDelta(p, c_2)$ would stay the same.

2. A voter who approves both c_1 and c_2, this time, $scoreDelta(p, c_1)$ and $scoreDelta(p, c_2)$ would increase equally by 1.

Proposition 3. *Having external candidate c_E and another internal one c_I, which is fully contained inside external's range.*

Let:

1. $K' = \{v \in \widehat{V} \mid v \in A(c_E) \wedge v \notin A(c_I)\}$.

2. $|K'| = scoreDelta(p, c_E) - scoreDelta(p, c_I)$.

3. $d(c) = score(c) - score(p) + 1$, *i.e., how many voters we need to delete for p to defeat c.*

Then $|K'| \geq 0$ is always true, and the size of set K' will be equal to the $d(c_E) - d(c_I)$

Proof. tbc. □

Special case - contained candidates

In the VCR model it is possible that some candidates have their interval fully contained in the other's interval. Every voter who would approve the smaller candidate would also approve the larger one, however not vice versa. For this case, our strategy of choosing voters to delete might seem suboptimal, at first glance.

The example from Figure 5.5 illustrates why we do not have to worry about this special case. Every contained voter will have as much or less total votes than the larger candidate, he/she might not even be a dangerous candidate at all. Now we see that in the process of defeating the larger candidate we will automatically defeat the smaller one.

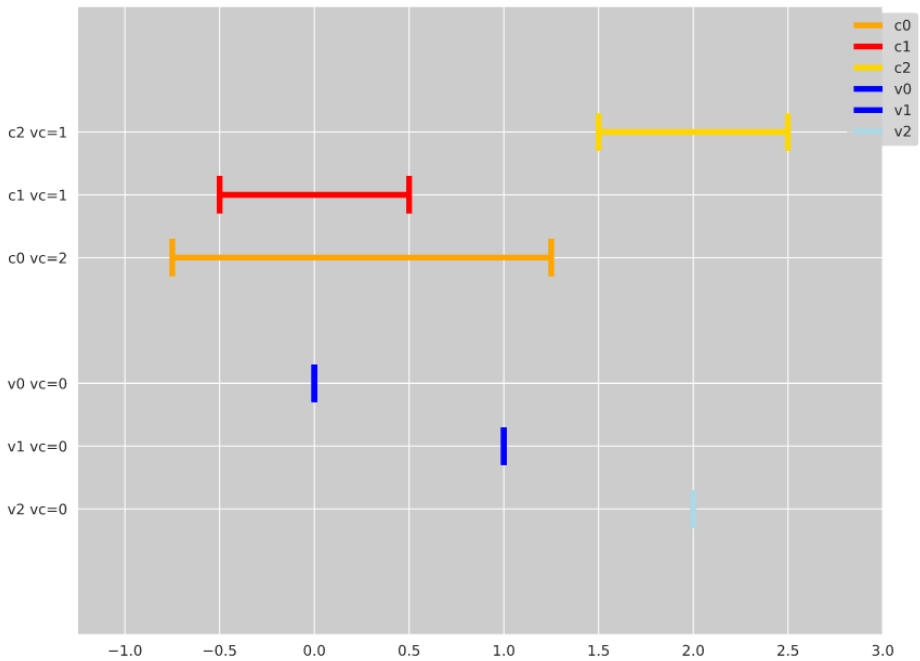

Figure 5.5 Temporary Image - Special Case

Choosing opponents

We consider only dangerous candidates (opponents), that is the ones with score higher or equal to the score of p. We process opponents one by one, starting with the one who has the leftmost beginning of the range, i.e., $min(b(c))$. After each iteration, we recompute the set of dangerous candidates. Thanks to this we are sure that all the steps we perform are necessary to find a solution (if it exists), hence optimal.

Voter deletion

Having currently processed opponent c_L, we consider only the voters who approve of him/her and do not approve of p, that is the voters whose range beginning is inside the candidate's range $\{v \in V' \mid b(v) \in [b(c_L), e(c_L)]\}$. We need to delete $scoreDelta(c_L, p) + 1$ voters to defeat the opponent. By picking the voters in order of the rightmost end of the range, i.e., $v_R = max(e(v))$, we try to maximize the number of other candidates who might be in the range $[b(v_R), e(v_R)]$. We don't have to worry about any candidate to the left, because of our leftmost candidate picking strategy, they are already defeated. This strategy ensures the solution is optimal if it exists.

5.3 AV-CC-AV for the VCR profiles

Our strategy is based on the solution from Magiera and Faliszewski (2014).

1. Add all $v \in W$ to the election
2. Is p the winner?
3. Try to delete $k' = |W| - k$ voters

Chapter 6

Evaluation

6.1 Conclusions

6.2 Future Work

Bibliography

Arrow, K. (1951). *Social Choice and Individual Values*. Wiley: New York.

Aziz, H. and Walsh, T. (2014). Justified representation in approval-based committee voting. *CoRR*, abs/1407.8269.

Barberà, S. and Moreno, B. (2011). Top monotonicity: A common root for single peakedness, single crossing and the median voter result. *Games and Economic Behavior*, 73(2):345–359.

Bartholdi, J. J., Tovey, C. A., and Trick, M. A. (1989). The computational difficulty of manipulating an election. *Social Choice and Welfare*, 6(3):227–241.

Bartholdi, J. J., Tovey, C. A., and Trick, M. A. (1992). How hard is it to control an election? *Mathematical and Computer Modelling*, 16(8):27–40.

Black, D. (1958). *The theory of committees and elections*. University Press, Cambridge.

Booth, K. S. and Lueker, G. S. (1976). Testing for the consecutive ones property, interval graphs, and graph planarity using pq-tree algorithms. *Journal of Computer and System Sciences*, 13(3):335 – 379.

Brams, S. and Fishburn, P. (1983). *Approval Voting*. Southgate Publishers.

Brandt, F., Brill, M., Hemaspaandra, E., and Hemaspaandra, L. A. (2015). Bypassing combinatorial protections: Polynomial-time algorithms for single-peaked electorates. *Journal of Artificial Intelligence Research*, 53:439–496.

Brandt, F., Conitzer, V., Endriss, U., Lang, J., and Procaccia, A. D., editors (2016). *Handbook of Computational Social Choice*. Cambridge University Press.

Elkind, E., Faliszewski, P., Skowron, P., and Slinko, A. (2017). Properties of multiwinner voting rules. *Soc. Choice Welf.*, 48(3):599–632.

Elkind, E. and Lackner, M. (2015). Structure in dichotomous preferences. *CoRR*, abs/1505.00341.

Elkind, E., Lackner, M., and Peters, D. (2016). Preference restrictions in computational social choice: Recent progress. In *Proceedings of the Twenty-Fifth International Joint Conference on Artificial Intelligence*, IJCAI'16, page 4062–4065. AAAI Press.

Faliszewski, P., Hemaspaandra, E., and Hemaspaandra, L. A. (2006). How hard is bribery in elections? *CoRR*, abs/cs/0608081.

Faliszewski, P., Hemaspaandra, E., Hemaspaandra, L. A., and Rothe, J. (2009). The shield that never was: Societies with single-peaked preferences are more open to manipulation and control. *CoRR*, abs/0909.3257.

Faliszewski, P. and Procaccia, A. D. (2010). Ai's war on manipulation: Are we winning? *AI Magazine*, 31(4):53–64.

Faliszewski, P., Skowron, P., Slink, A., and Talmon, N. (2017). *Multiwinner Voting: A New Challenge for Social Choice Theory*. AI Access.

Fitzsimmons, Z. and Lackner, M. (2019). Incomplete preferences in single-peaked electorates. *CoRR*, abs/1907.00752.

Godziszewski, M., Batko, P., Skowron, P., and Faliszewski, P. (2021). An analysis of approval-based committee rules for 2d-euclidean elections.

Hemaspaandra, E., Hemaspaandra, L. A., and Rothe, J. (2009). Hybrid elections broaden complexity-theoretic resistance to control. *Mathematical Logic Quarterly*, 55(4):397–424.

Lackner, M. and Skowron, P. (2020). Approval-based committee voting: Axioms, algorithms, and applications.

Liu, H. and Guo, J. (2016). *Parameterized Complexity of Winner Determination in Minimax Committee Elections*, page 341–349. AAMAS '16. International Foundation for Autonomous Agents and Multiagent Systems.

Magiera, K. and Faliszewski, P. (2014). How hard is control in single-crossing elections? *Frontiers in Artificial Intelligence and Applications*, 263:579–584.

Mirrlees, J. A. (1971). An Exploration in the Theory of Optimum Income Taxation. *Review of Economic Studies*, 38(2):175–208.

Roberts, K. W. (1977). Voting over income tax schedules. *Journal of Public Economics*, 8(3):329–340.

Skowron, P., Yu, L., Faliszewski, P., and Elkind, E. (2015). The complexity of fully proportional representation for single-crossing electorates. *Theoretical Computer Science*, 569:43–57.

Walsh, T. (2007). Uncertainty in preference elicitation and aggregation. In *Proceedings of the 22nd National Conference on Artificial Intelligence - Volume 1*, AAAI'07, page 3–8. AAAI Press.